Dell PowerFlex Appliance with PowerFlex 5.x

Architecture Overview

June 2026
Rev. 2.0

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1 Introduction

This guide describes the high-level design of PowerFlex appliance.

This guide is applicable for PowerFlex software 5.0 and later managed by PowerFlex Manager 5.0 and later. For PowerFlex Manager 5.1.x managing PowerFlex software 4.5.5 and later, see the PowerFlex rack or PowerFlex appliance with PowerFlex 4.x documentation.

The target audience for this document includes customers, sales engineers, field consultants, and advanced services specialists who want to deploy a high-performance, scalable, and flexible infrastructure using PowerFlex appliance.

PowerFlex appliance architecture is based on Dell PowerEdge servers; Cisco Nexus switches, or Dell PowerSwitch switches, and PowerFlex software-defined storage. PowerFlex Manager provides the management and orchestration functionality for PowerFlex appliance.

This Dell Technologies product uses an embedded Linux distribution that is licensed to Dell. This embedded Linux is only authorized to be used to operate the Dell product and only the Dell product.

For additional information on the PowerFlex architecture, refer to the *Dell PowerFlex and PowerFlex Manager Technical Overview*. This includes the PowerFlex limit table.

For additional PowerFlex appliance documentation, go to PowerFlex appliance technical documentation.

Overview

PowerFlex appliance is an engineered system designed to meet modern data center needs. You can deploy two layer (separate compute-only and storage-only nodes), storage-only, or hybrid combinations.

PowerFlex appliance is a modular software-defined compute and storage platform that enables linear performance with scale and flexible deployment options for next-generation cloud applications and mixed workloads. The scale-out architecture of the PowerFlex appliance enables you to add PowerFlex nodes with various CPU, memory, and drive configuration options to meet the business need. PowerFlex appliance is designed for deployments involving large numbers of virtualized and bare metal workloads. PowerFlex appliance is built-in with protection architecture based on erasure coding, enabling a two disk and two node failure while maintaining data availability, which means architecture is resilient to two concurrent device or node failures.

PowerFlex appliance has many advantages:

- Engineered system with automated end-to-end life cycle management using PowerFlex Manager
- Integrated architecture with choice of the following network topologies to meet scale and performance business needs:
  - Access and aggregation
  - Leaf-spine
- Vendor choice on network hardware or customer preferred: Cisco Nexus or Dell PowerSwitch switches
- Multiple PowerFlex node types and node configurations options to meet compute and storage needs:
  - PowerFlex storage-only nodes
  - PowerFlex compute-only nodes
- Flexible compute and storage resources deployment options such as:
  - **Two layer** - compute and storage deployed in separate chassis allowing independent scale of compute and storage resources
  - **Storage only** - only storage resources are part of PowerFlex appliance, compute resides outside the system boundaries
  - **Hybrid** - combination of two or more of the above deployment options
- Highly available management and orchestration control plane that runs on a dedicated three or more physical nodes cluster
  - Cost effective management and orchestration that runs on a single physical PowerFlex management node
  - Use of your existing servers for management and orchestration
- Supports managed VMware ESXi and bare metal Linux (unmanaged) options as compute-only
- Software and hardware-based data at rest encryption (D@RE) options
  - Dell CloudLink
  - Optional SEDs
- Supports 25 GbE or 100 GbE port bandwidth for backend connectivity

- Dual network environment using your existing software-defined network (SDN), such as Cisco ACI (optional) or VMware NSX
- Supports block storage
- Support for management aggregation switches (optional)
- Built-in component level redundancy to ensure data availability
- Self-healing architecture with integrated call home feature
- Storage only option allows external compute resources to access data in the PowerFlex appliance
- PowerFlex nodes support:
  - SSD
  - NVMe technologies
  - Software-defined persistent memory (SPDM) for PowerFlex R760 and R660 nodes

  NOTE: Persistent memory (software-defined persistent memory/NVDIMM) is not mandatory for inline data compression. PMEM is used for write cache and metadata persistency in the protection domain with erasure coding.

- Supports multi-VLAN or multi-subnet for the same network type, other than data networks and NSX overlay
- Supports non-root user, SSH key pairs and LDAP users for PowerFlex administration functions for security
- PowerFlex Manager supports dynamic licensing and key-based licensing. Dynamic licensing automates license management activities, such as registrations, capacity expansions, and subscription renewals. Key-based licensing allows you to upload a single file to PowerFlex Manager that includes both PowerFlex Manager and PowerFlex licensing information. See the *Dell PowerFlex Family Licensing Guide* for more information.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
device gateway target	DGWT
management virtual machine	MVM
metadata manager	MDM
PowerFlex data server	PDS
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data target	SDT

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

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Revision history

Table 2. Revisions

Date	Document revision	Description of changes
June 2026	2.0	Added support for: • 4+2 protection scheme • PowerFlex R670/R770/R6725/ R7725 compute-only nodes • Dynamic licensing
November 2025	1.1	Technical update
October 2025	1.0	Initial release

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Architecture considerations

PowerFlex appliance is a modular platform that enables extreme scalability and flexibility for next-generation cloud applications and mixed workloads.

System components

PowerFlex appliance contains compute, network, software-defined storage, virtualization, management and orchestration control plane.

Software components

Table 3. Software components

Software	Function	Description
PowerFlex Manager	Management and orchestration	PowerFlex Manager is the management layer of the PowerFlex system. It provides the user interface for the block storage (PowerFlex software) and lifecycle management of the hardware components in the PowerFlex appliance.
PowerFlex	Software-defined storage	PowerFlex software, provides block services to the PowerFlex system and is the software-defined storage layer that forms the core of the offer.
VMware vSphere	Virtualization	• VMware ESXi: This is the default supported hypervisor for PowerFlex compute-only nodes. • VMware vCenter Server Appliance (vCSA): The VMware vCSA provides management services to the VMware compute-only environment.
Secure connect gateway	Call home	Secure connect gateway is an enterprise monitoring technology. It monitors your devices and proactively detects hardware issues that may occur. It automates support request creation for issues that are detected on the monitored devices. **NOTE:** Secure connect gateway automatically collects the telemetry that is required to troubleshoot the issue that is detected. The collected telemetry helps technical support to provide a proactive and personalized support experience. Secure connect gateway can be set up with policy manager.
Policy manager	Device access management	Policy manager is a device access management technology that is delivered as a virtual appliance.
Dell CloudLink (optional)	Software encryption and key management	CloudLink is an optional component that provides key management for self-encrypting drives, and software data-at-rest encryption for non-self-encrypting units.

Hardware components

Table 4. Hardware components

Resource	Vendor	Components
Storage	Dell	PowerFlex nodes • PowerEdge R660/R760/R650/R750 servers with PowerFlex software-defined storage
Compute	Dell	PowerFlex nodes:

Table 4. Hardware components (continued)

Resource	Vendor	Components
		• PowerEdge R670/R770/R6725/R7725 servers or PowerEdge R660/R760/R6625/R7625/R860 servers or PowerEdge R650/R750/R6525/R7525 servers for PowerFlex compute-only nodes 4 x 25 Gbe or 4 x 100 Gbe NIC options
Network	Cisco	Management switch option: • Cisco Nexus 9348GC-FX3 Management aggregation switch option: • Cisco Nexus 93180YC-FX3 Aggregation switch option: • Cisco Nexus 9336C-FX2 Access switch options: • Cisco Nexus 93240YC-FX2 • Cisco Nexus 93180YC-FX3 Spine switch options: • Cisco Nexus 9336C-FX2 • Cisco Nexus 9364C-GX Leaf switch options: • Cisco Nexus 93240YC-FX2 • Cisco Nexus 9336C-FX2 • Cisco Nexus 9364C-GX Border-leaf switch option: • Cisco Nexus 9336C-FX2 Customer access switch (for optional ACI connectivity using dual network) • Cisco Nexus 93240YC-FX2
	Dell	Management switch option: • Dell PowerSwitch S3248T-ON • Dell PowerSwitch S4148T-ON Aggregation switch option: • Dell PowerSwitch S5232F-ON Access switch option: • Dell PowerSwitch S5248F-ON
		SONiC networking Management switch option: • Dell PowerSwitch S3248T-ON Aggregation switch option: • Dell PowerSwitch Z9432F-ON Access switch option: • Dell PowerSwitch S5448F-ON Management aggregation switch option: • Dell PowerSwitch S5448F-ON
Management control plane	Dell	PowerFlex management cluster: • PowerEdge R660 or PowerEdge R650 servers custom configuration

Key architecture considerations

Flexible network architecture is a key value proposition of PowerFlex appliance.

In addition to vendor (Cisco Nexus, Dell PowerSwitch or customer preferred switches), PowerFlex appliance architecture offers the following network topologies to meet your business needs:

- Access and aggregation
- Leaf-spine

PowerFlex appliance also offers the ability to support both hardware enabled software-defined networking (Cisco Application Centric Infrastructure) and native software-defined networking (VMware NSX).

PowerFlex appliance requires a minimum five nodes for storage-only configuration.

You can deploy PowerFlex compute-only nodes or PowerFlex storage-only nodes using one or more of the following resource deployment options:

- Storage-only deployment
- Two-layer deployment with disaggregated compute and storage only
- Hybrid deployment as a combination of above

PowerFlex Manager allows you to specify a non-root user when configuring a template for compute-only or storage-only.

PowerFlex appliance can be deployed with either full network automation or partial network automation. With full network automation, PowerFlex Manager configures the node facing ports on the customer network switches, if they are Dell or Cisco Nexus supported switches outlined in *System components*. Partial network automation is used when you have customer network switches that are not supported by Dell. In this case, you are responsible for configuring the node facing ports along with the rest of your network. The PowerFlex nodes can be fully managed by PowerFlex Manager in either full network automation or partial network automation mode.

Network architecture

PowerFlex appliance is available as two different network architectures to meet performance and scaling requirements.

Access and aggregation architecture

PowerFlex appliance uses an access and aggregation architecture to meet performance and scaling requirements.

The minimum node count is 5 (2+2), 7 (4+2), or 11 (8+2).

The following figure shows the logical layout of the access and aggregation architecture:

NOTE: There is an additional 10/25 Gb link from the PowerFlex management nodes to the out-of-band management switch.

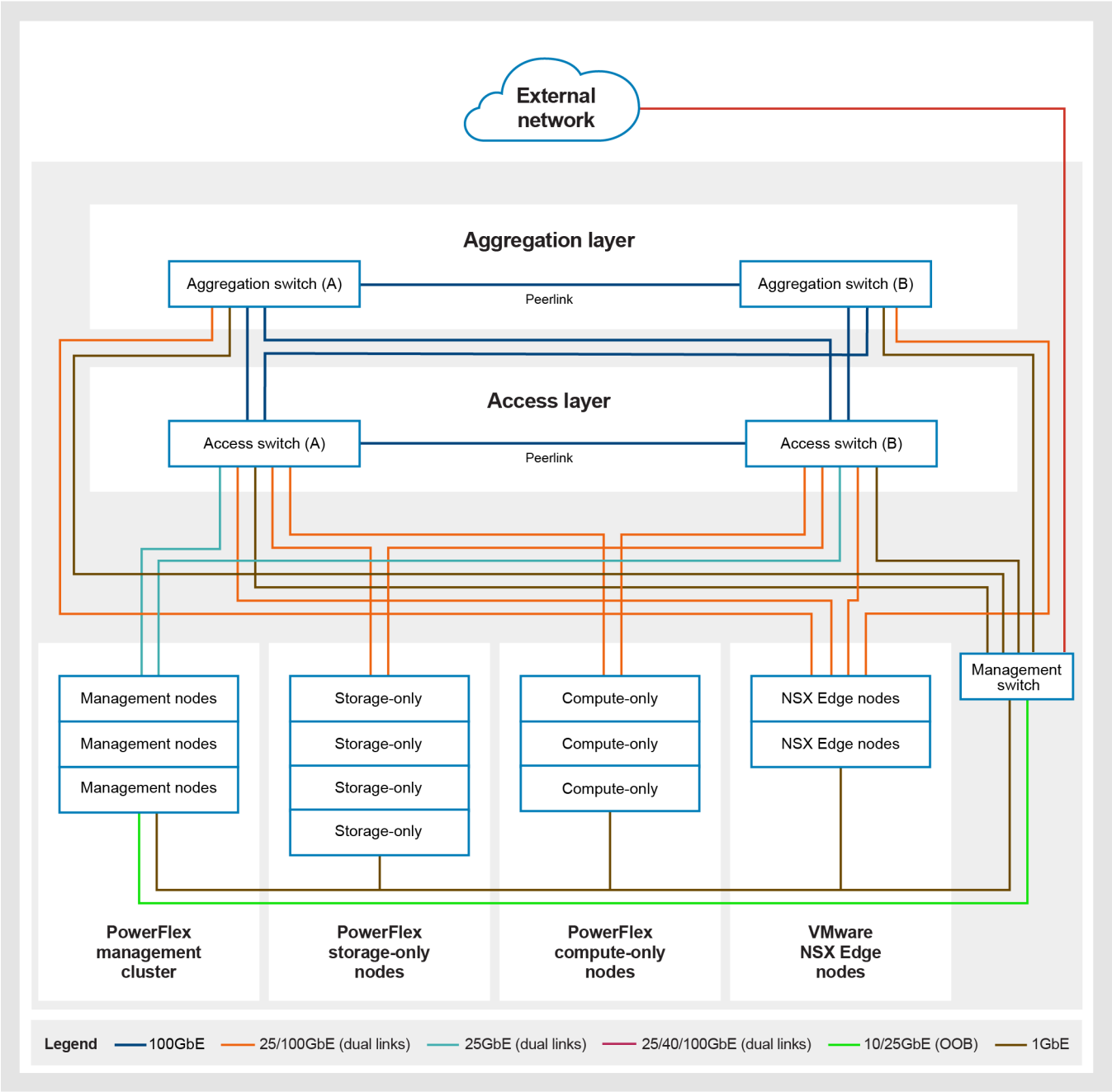


Figure 1. Logical layout of the access and aggregation architecture

The following figure shows the logical layout of the access and aggregation with management-aggregation switch architecture:

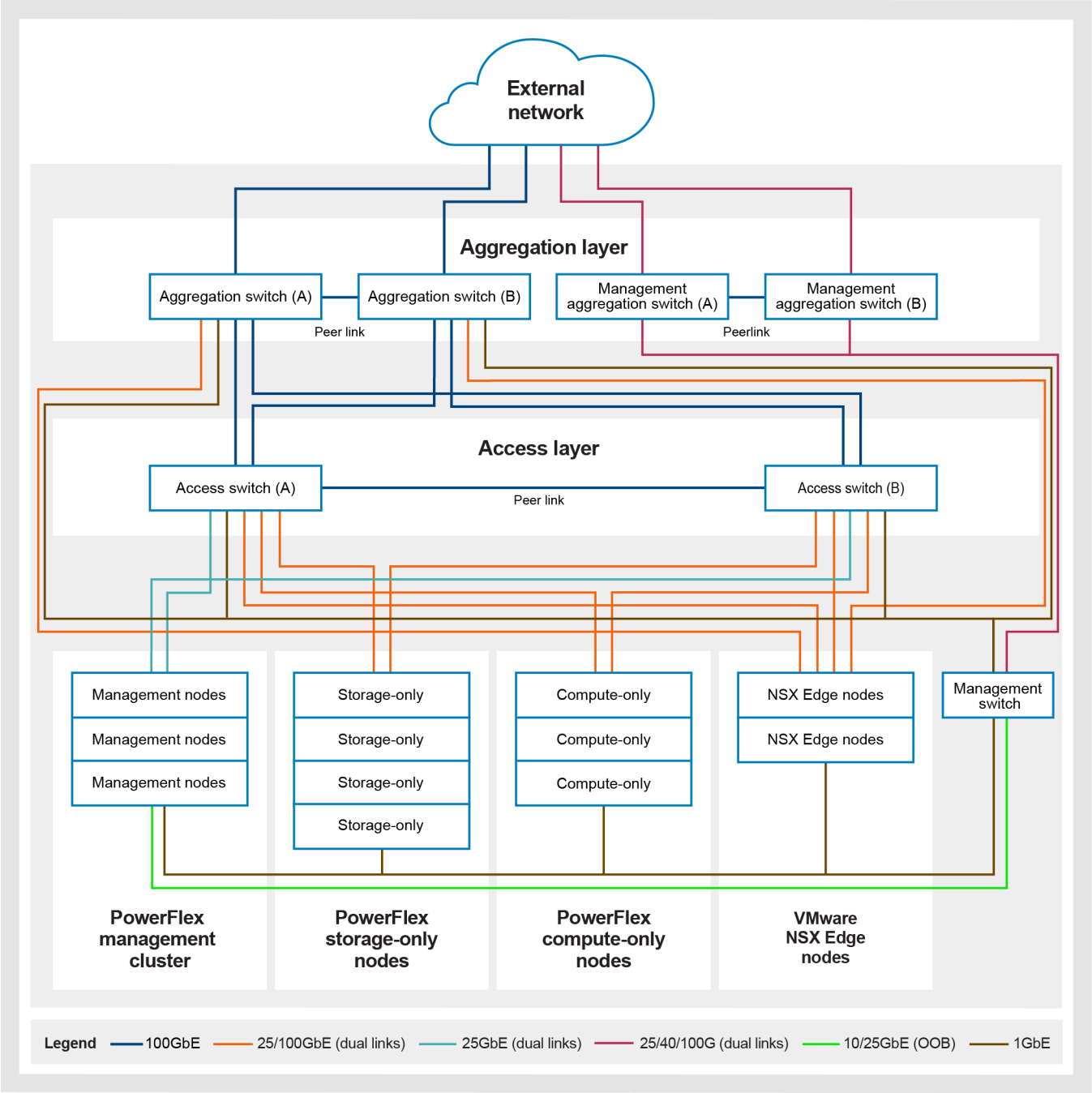


Figure 2. Logical layout of the access and aggregation with management-aggregation switch architecture

NOTE: SONiC network configurations do not support VMware NSX deployments.

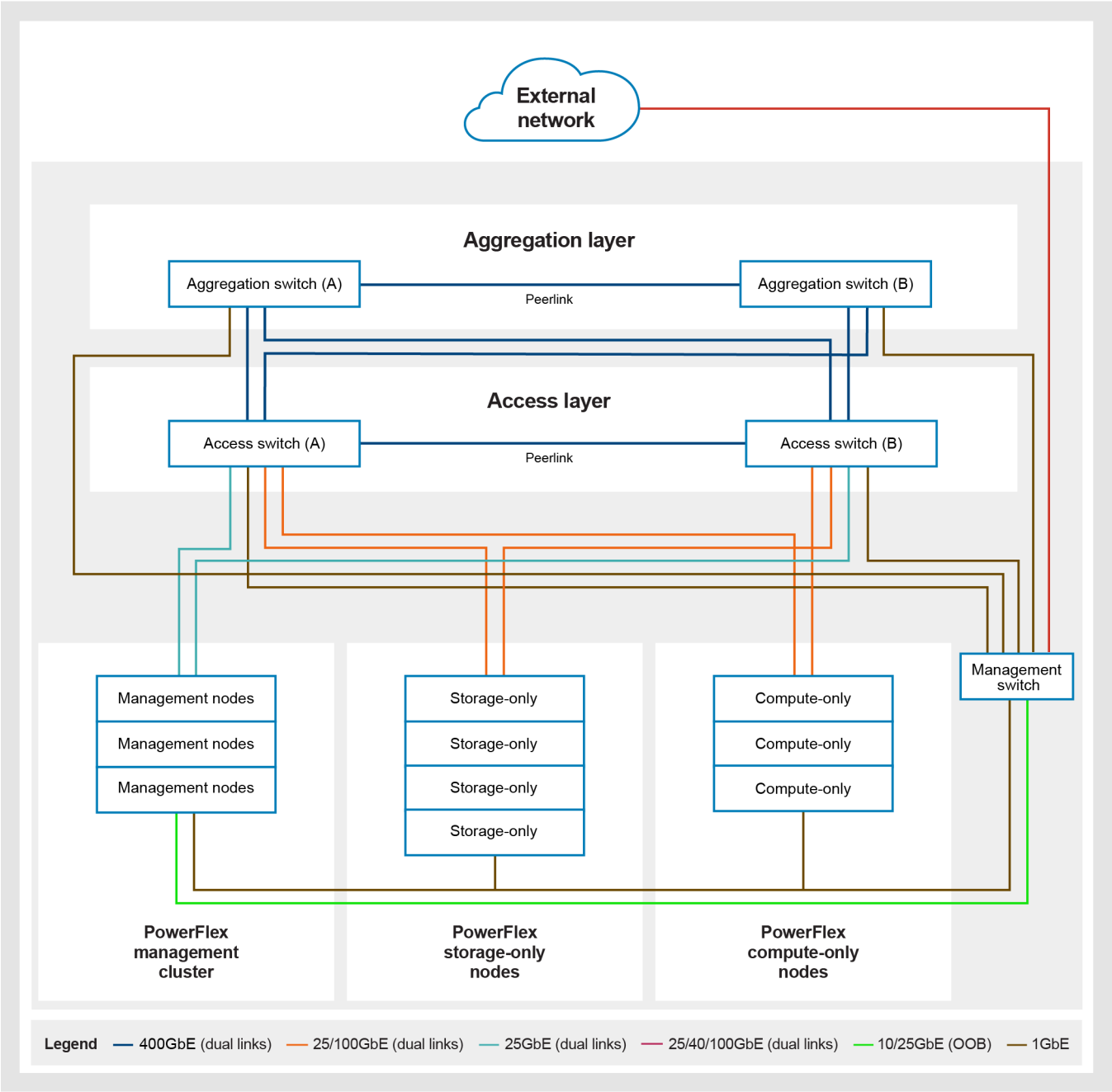


Figure 3. Overview of access-aggregation switches for SONiC networking

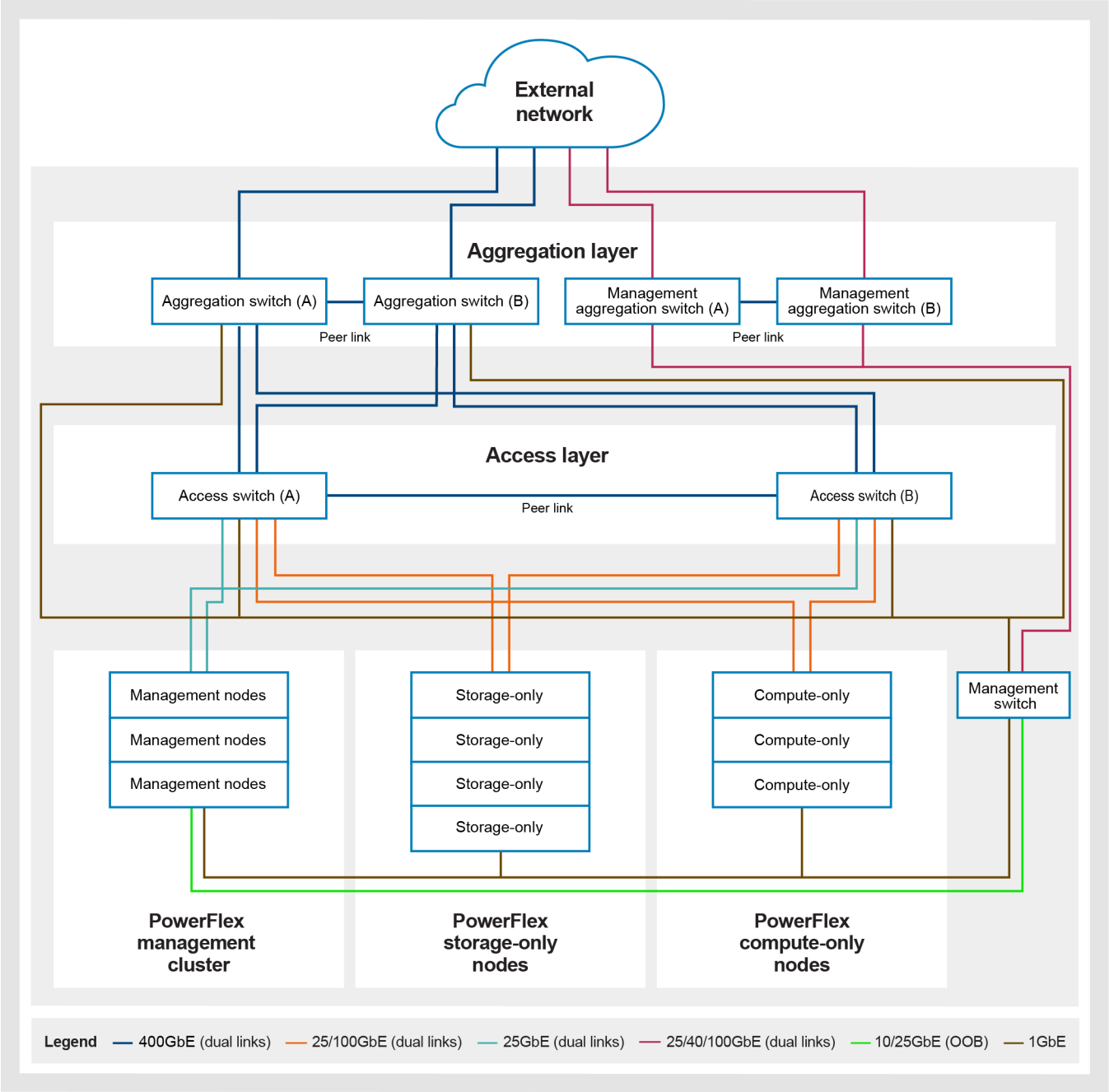


Figure 4. Overview of access-aggregation block with management-aggregation switches for SONiC networking

Leaf-spine architecture

Leaf-spine architecture is 100 GbE connectivity from every spine to every leaf. Network switches provide two options to support leaf-to-node connectivity.

The minimum node count is 5 (2+2), 7 (4+2), or 11 (8+2).

The following diagram shows the logical layout of the leaf-spine architecture:

NOTE: There is an additional 10/25 Gb link from the PowerFlex management nodes to the out-of-band management switch.

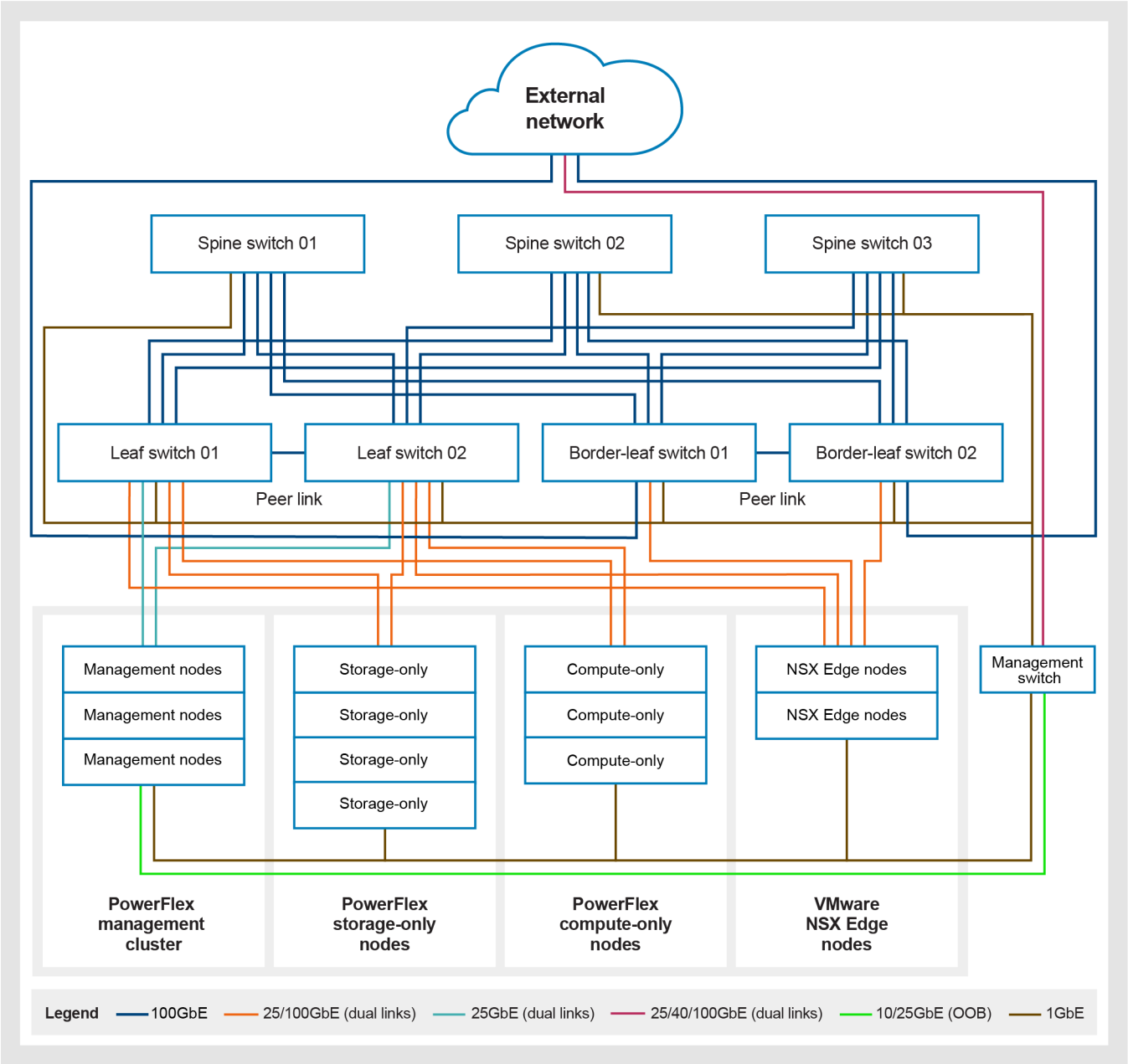


Figure 5. Logical layout of the leaf-spine architecture

The following diagram shows the logical layout of the leaf-spine with management-aggregation switch architecture:

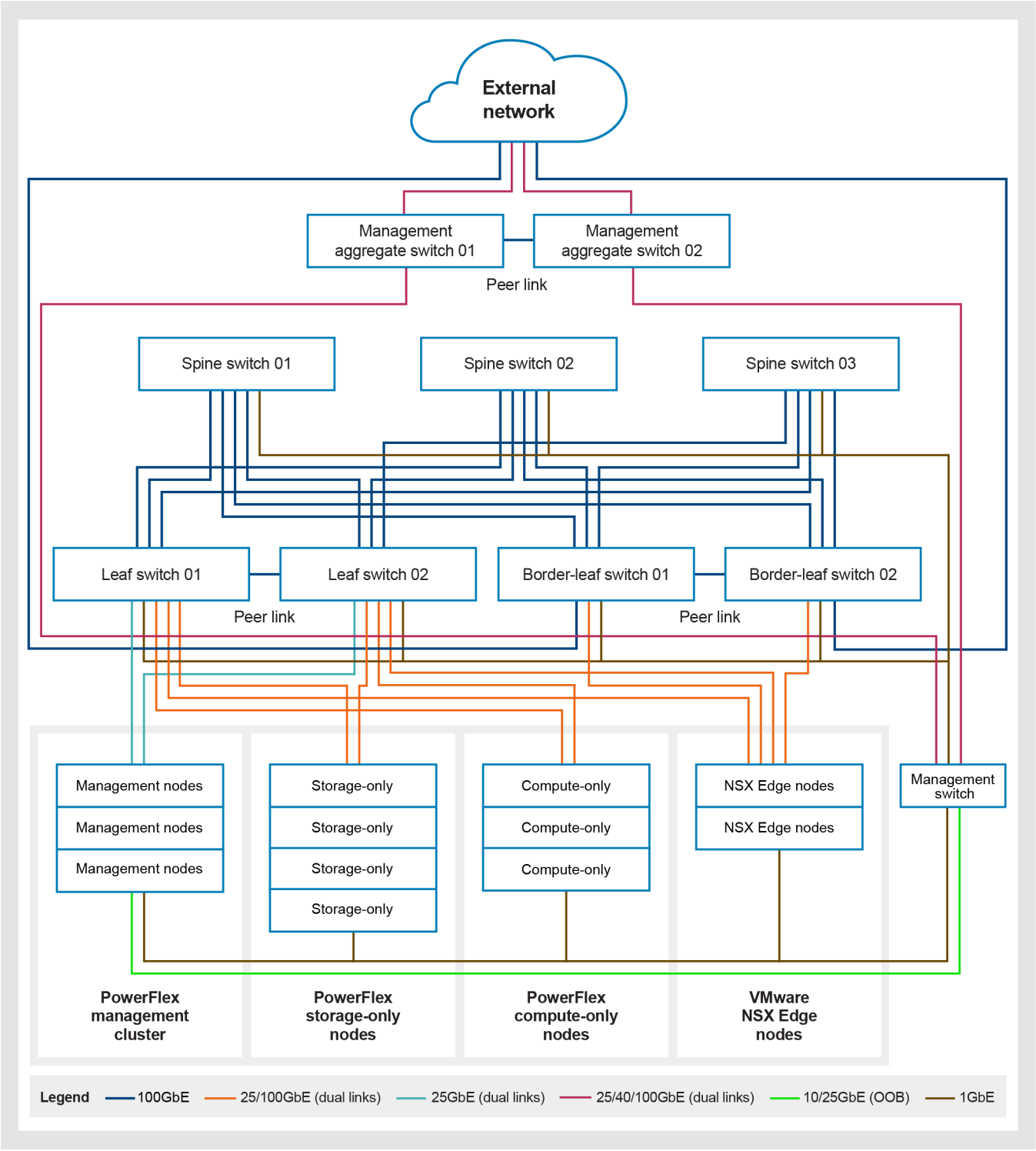


Figure 6. Logical layout of the leaf-spine with management-aggregation switch architecture

NOTE: The minimum number of supported PowerFlex storage-only nodes is five nodes.

VMware NSX ready architecture

The VMware NSX ready deployment is a variation of standard deployment that includes PowerFlex compute-only nodes and an additional VMware NSX Edge node cluster deployment.

The following table contains VMware NSX ready requirements for each VMware vSphere cluster:

Table 5. VMware NSX ready requirements

VMware vSphere cluster	Requirements for PowerFlex appliance
Management cluster	Created as a single VMware vSphere cluster and added within the PowerFlex data center. • The three NSX Manager VMs run in this cluster.
NSX Edge cluster	• Created as a single VMware vSphere cluster and added in the PowerFlex vCenter Server. • HA enabled and DRS disabled at the VMware vSphere cluster level, if running the vSAN storage option. • Deployed across multiple physical racks (split between two racks). • By default, local storage with RAID 1+0 is enabled. However, vSAN is supported if you or VMware services choose this option. • Includes NSX Edge Gateway VMs, deployed at the customer site, that provide logical routing. • At minimum, four of the six NIC interfaces that are used for transport and external edge traffic must be configured as an individual trunk. The other two NIC interfaces that are used for VMware ESXi management or vSAN traffic are configured with link aggregation control protocol (LACP) enabled vPC. **NOTE:** Do not deploy non-NSX Edge workloads in the NSX Edge VMware vSphere cluster.
Compute cluster	• On a compute cluster, the front-end distributed switch must be configured as trunk ports. The switch carries overlay NSX ready transport traffic, and VMware does not recommend using LACP. • Includes the customer production VMs to run on the NSX transport overlay VMkernel.

PowerFlex storage-only deployment

A storage-only deployment has a base configuration that is a minimum set of PowerFlex storage-only nodes and fixed network resources.

Within the base configuration, you can customize the following hardware aspects:

Table 6. Storage-only deployment requirements

Hardware	Minimum set
Network	• One management switch • One pair of access or leaf switches • One pair of aggregation switches or three spine switches • One pair of management aggregation switches (optional) • One pair of border-leaf switches **NOTE:** Only in a leaf-spine configuration.
Storage	At least five PowerFlex storage-only nodes. Persistent memory using hardware components such as software-defined persistent memory (SDPM) or NVDIMMs is no longer mandatory for PowerFlex inline data compression. PowerFlex uses persistent memory for write cache and metadata persistency within the protection domain. Persistent memory also provides improved performance and reduced drive wear.
Management	• PowerFlex management cluster (KVM) • Single ESXi PowerFlex management node • Customer-provided infrastructure for PowerFlex management (ESXi) • Customer-provided infrastructure for PowerFlex management (KVM/Linux)

PowerFlex two-layer deployment

A two-layer deployment has a base configuration that is similar to a PowerFlex storage-only node deployment, but adds a minimum set of PowerFlex compute-only nodes. The minimum set of PowerFlex storage-only nodes and fixed network resources are also required.

Within the base configuration, you can customize the following hardware aspects:

Table 7. PowerFlex two-layer deployment requirements

Hardware	Minimum set
Compute	At least three PowerFlex compute-only nodes.
Network	• One customer provided management switch • One pair of access or leaf switches (Dell PowerSwitch switches or customer provided switches) • One pair of border-leaf switches **NOTE:** Only in a leaf-spine configuration.
Storage	At least five PowerFlex storage-only nodes. Software-defined SAN storage (uses local disks to build a PowerFlex storage pool). Persistent memory using hardware components such as software-defined persistent memory (SDPM) or NVDIMMs is no longer mandatory for PowerFlex inline data compression. PowerFlex uses persistent memory for write cache and metadata persistency within the protection domain. Persistent memory also provides improved performance and reduced drive wear.
Management	• PowerFlex management cluster (KVM) • Single ESXi PowerFlex management node • Customer-provided infrastructure for PowerFlex management (ESXi) • Customer-provided infrastructure for PowerFlex management (KVM/Linux)

VMware NSX Edge node deployment

The NSX ready deployment consists of the specification provided in the following table:

Table 8. VMware NSX Edge node deployment requirements

Hardware	Minimum set
Compute	VMware NSX transport is configured on PowerFlex compute-only nodes.
Network	• Supports either a traditional Ethernet architecture (Cisco Nexus or Dell PowerSwitch) or leaf-spine topology (Cisco Nexus). • By default, the NSX Edge physical nodes connect directly to either the aggregation or border leaf switches, depending on the network topology. If there is a limitation because of port capacity or cable distance, the management and transport connections (not Edge/BGP uplinks) are relocated from the aggregation or border leaf switches to the access or leaf switches.
Storage	• NSX Edge nodes can run in either a local RAID1+0 storage (recommended) or a VMware vSAN storage solution. • NSX Managers will run on the general shared datastores provided by PowerFlex within the PowerFlex management cluster. • PowerFlex storage-only nodes are not supported as an NSX transport node.
Management	• If you have an existing PowerFlex management cluster (KVM), you must have the PowerFlex management cluster (KVM) compute-only extension (ESXi-based) to support the NSX deployment. • You can bring your own VMware vCenter for the NSX deployment.
NSX Edge	• A minimum of two NSX Edge nodes, if using local RAID storage option. • A minimum of four NSX Edge nodes, if using vSAN storage option.

Table 8. VMware NSX Edge node deployment requirements (continued)

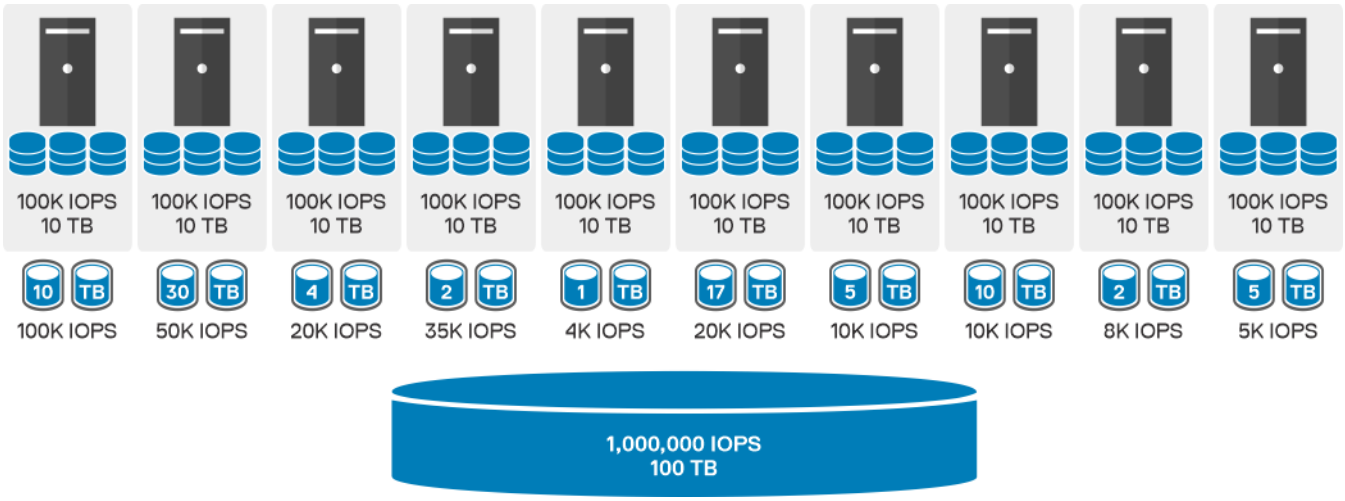
Hardware	Minimum set
	Each NSX Edge node uses three dual-port 25 Gb cards to connect to either the border leaf or aggregation switches. At minimum, four of the six NIC interfaces that are used for transport and external edge traffic must be configured as an individual trunk. The other two NIC interfaces that are used for VMware ESXi management or vSAN traffic are configured with link aggregation control protocol (LACP) enabled vPC. **NOTE:** Do not deploy non-NSX Edge workloads in the NSX Edge VMware vSphere cluster.

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PowerFlex software-defined storage architecture

PowerFlex applies the principles of server virtualization to standard x86 servers with local disks, creating high-performance, sharable pools of block storage. PowerFlex abstracts the local storage contained within each server.

PowerFlex pools all the storage resources together. In the following figure, there is a global pool of 1 million IOPS and 100 terabytes, instead of having 100K IOPS and 10 terabytes available in each server. The applications are not constrained by what is within the local server, these resources are shared across the entire cluster.



PowerFlex automatically maintains balance across all resources, supporting application needs. Storage and/or compute can be added dynamically with no downtime or impact to applications because PowerFlex seamlessly balances the available resources. This enables data center operation in the most efficient and cost-effective way possible, regardless of organization size.

PowerFlex components

PowerFlex data server (PDS)

The PowerFlex data server is responsible for processing host I/Os. It is installed on every PowerFlex node that contributes storage to the system. The PowerFlex data server abstracts local storage, maintains device groups, storage pools, and presents volume to the storage data client.

Device gateway target (DGWT)

The device gateway target is accountable for placing data on the drives using erasure coding protection schemes. It is installed on every PowerFlex node that contributes storage to the system.

Metadata manager (MDM)

The metadata manager software installs on three or five PowerFlex nodes and forms a cluster that supervises the operations of the entire cluster and its parts, while staying outside of the data path itself. The metadata manager hands out instructions to each storage data client and storage data server about its role and how to perform it, giving each component the information it needs. The metadata manager:

- Oversees storage cluster configurations, monitoring, rebalances, and rebuilds.
- Is a highly available, independent cluster installed on three or five different PowerFlex nodes.
- Sits outside the data path.

Lightweight Installation Agent (LIA)

The lightweight installation agent is a required component for many maintenance operations. During initial installation, the installation manager establishes trust with the LIA, which facilitates the operation. The LIA is installed on every PowerFlex node that is part of the PowerFlex system.

ActiveMQ

ActiveMQ is used for sending MDM events to PowerFlex Manager. ActiveMQ is installed only on PowerFlex nodes where primary and secondary MDM components are running.

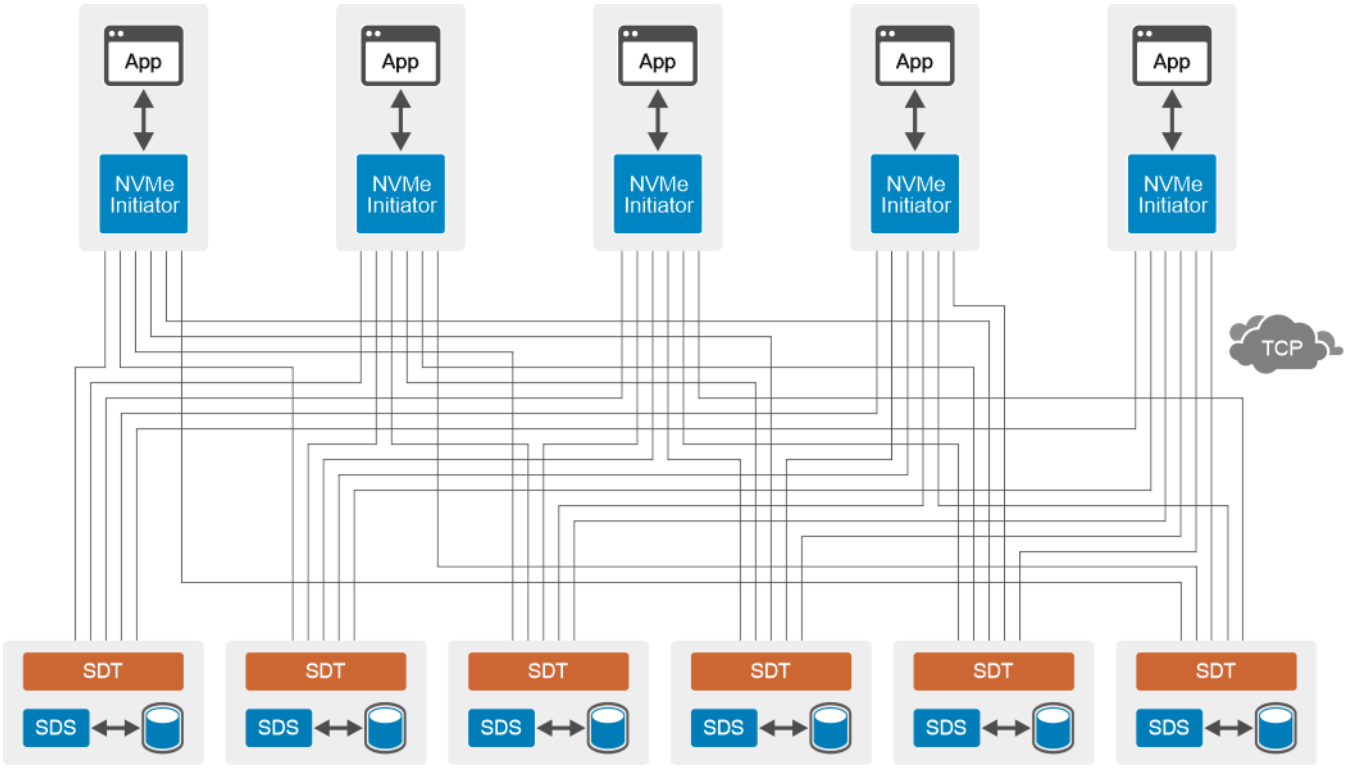
Storage data client (SDC)

The storage data client is installed on PowerFlex nodes that consume the system storage volumes. The volumes data and copies are spread evenly across the nodes and drives that comprise the pool. The storage data client communicates over multiple pathways to all the nodes. In this multi-point peer-to-peer fashion, it reads and writes data to and from all points simultaneously, eliminating bottlenecks and quickly routing around failed paths. The storage data client:

- Provides front-end volume access to applications and file systems.
- Is installed on servers consuming storage.
- Maintains peer-to-peer connections to every storage node managing a pool of storage.

Storage data target (SDT)

The storage data target (SDT) is installed with the storage data server to connect compute/application clients to storage using NVMe over TCP. NVMe over TCP front-end capability allows you to use agentless solution (no storage data client), providing more flexible options for operating systems where storage data client is not supported and reducing the operational complexity of deploying and maintaining the host agent.



Storage nodes

Storage nodes are physical or virtual servers that run your chosen PowerFlex operating system and store PowerFlex information. Storage nodes support the actions that affect software components running on the operating system and the storage and PMEM devices that are installed on the physical node.

You can add a storage node to your PowerFlex system to scale up your storage without having to add compute resources. You can distribute data across multiple storage nodes over a fault-tolerant, high-speed network. If one storage node fails, the PowerFlex system can continue operating with minimal impact and continue to protect against data loss. Storage nodes are accessed and managed under the **Lifecycle** tab in PowerFlex Manager.

Each storage node is installed on a server that contributes storage devices and optional persistent memory (PMEM), or an AWS EC2 instance or Microsoft Azure virtual machine that contributes storage devices only. Storage nodes contain the following components:

- PowerFlex data server (PDS): The frontend application that is responsible for processing host I/Os.
- Device gateway target (DGWT): The component that is accountable for placing data on the storage drives using erasure coding protection schemes.

The minimum number of storage nodes that are deployed in the cluster depends on the erasure coding protection scheme that you are using:

- 2+2 protection scheme: five storage nodes.
- 4+2 protection scheme: seven nodes
- 8+2 protection scheme: 11 storage nodes.

Storage nodes can be temporarily removed from live systems for maintenance and troubleshooting purposes, but cannot be permanently removed.

Protection domains

A protection domain is a logical entity that consists of a group of storage nodes that back each other up.

Each storage node belongs to one protection domain, so each protection domain contains a unique set of storage nodes that are defined to behave as an independent system. All storage objects (volumes, storage pools, and the like) are contained within the protection domain.

There are no logical connections between nodes in different protection domains.

The following image displays a PowerFlex deployment with three protection domains. Protection domain 1 is configured in an 8+2 protection scheme, and protections domains 2 and 3 are configured in a 2+2 protection scheme. The protection domains all share the same MDM cluster (configured on protection domain 1), as required for a single deployment.

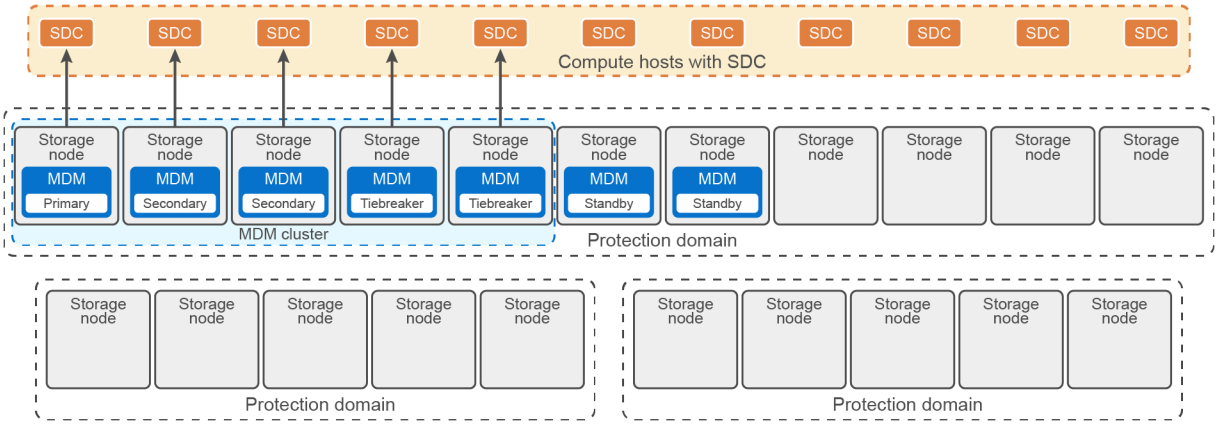


Figure 7. Multiple protection domains in a single PowerFlex deployment

Each protection domain supports up to 128 storage nodes. This enables:

- Optimal performance
- A reduction of theoretical *mean time between failure* issues
- The ability to sustain multiple failures in different protection domains

Protection domains are configured exclusively to support the designated protection scheme. You can add protection domains during installation and modify existing protection domains after installation.

Device groups

A device group is a set of physical devices within a protection domain. Device groups represent the raw capacity of a group of devices. A device group is responsible for grouping together devices of a similar type, which makes device management easier. Device groups can also be used for tiering.

PowerFlex supports two different types of device groups: storage devices (SDPMs) and persistent memory devices (NVDIMMs). Each device group must contain storage devices on a minimum of four separate nodes.

Device groups containing storage devices only support SSD and PCI SSD (NVMe) drive types. Dell Technologies recommends that only one drive type be configured for each device group for optimal performance.

Device groups are physically bound by the amount of physical storage in each group. Device groups must be uniform in terms of number of devices, with the same number of devices in each device group. You can view device groups under the **Block** tab in PowerFlex Manager.

NOTE: Dell Technologies does not recommend mixing different types of SSDs and NVMe drives in a device group. PowerFlex does not perform optimally if there are differences in size or type.

Storage pools

A storage pool is a group of volumes. A storage pool allocates capacity from a device group to create the volumes, and sets the base erasure coding protection scheme, which is either 2+2, 4+2, or 8+2. Only one storage pool is configured for each device group and its protection domain.

The protection scheme that is selected during deployment cannot be changed to an alternate protection scheme after deployment. For example, customers starting with a 2+2 (five-node) protection scheme cannot move workloads to a 4+2 (seven-node) or an 8+2 (11-node) protection scheme. Restriping is not supported. Restriping is the process of changing the geometry, such as from 2+2 to 8+2.

Storage pools can be created from nodes with different device types. The mixing of SAS, SATA, and NVMe is permitted, but mixing device types reduces overall storage performance to that of the slowest device. Individual SDSs must have drives of a single type, but mixing those nodes together is allowed.

PowerFlex enables data compression by default, but it can be disabled during storage pool creation. You can also optionally define the over provisioning ratio for the storage pool.

Over-provisioning control

Over-provisioning is an optional control that allows you to set a limit that is defined as a percentage factor of the physical size of the storage pool. For example, if the storage pool is 10 TB and the over-provisioning factor is 200%, you can create thin volumes up to 20 TB. The default over-provisioning value is unlimited.

When over-provisioning is set, the system enforces the over-provisioning limit. If the sum total of the sizes of all volumes and snapshots created for the storage pool reaches the over-provisioning limit, no new volumes or snapshots can be created.

Volumes

Volumes are a portion of the storage pool capacity, evenly distributed across storage devices. They appear as a single object in PowerFlex.

Volumes must be mapped to hosts before an application or operating system can access them.

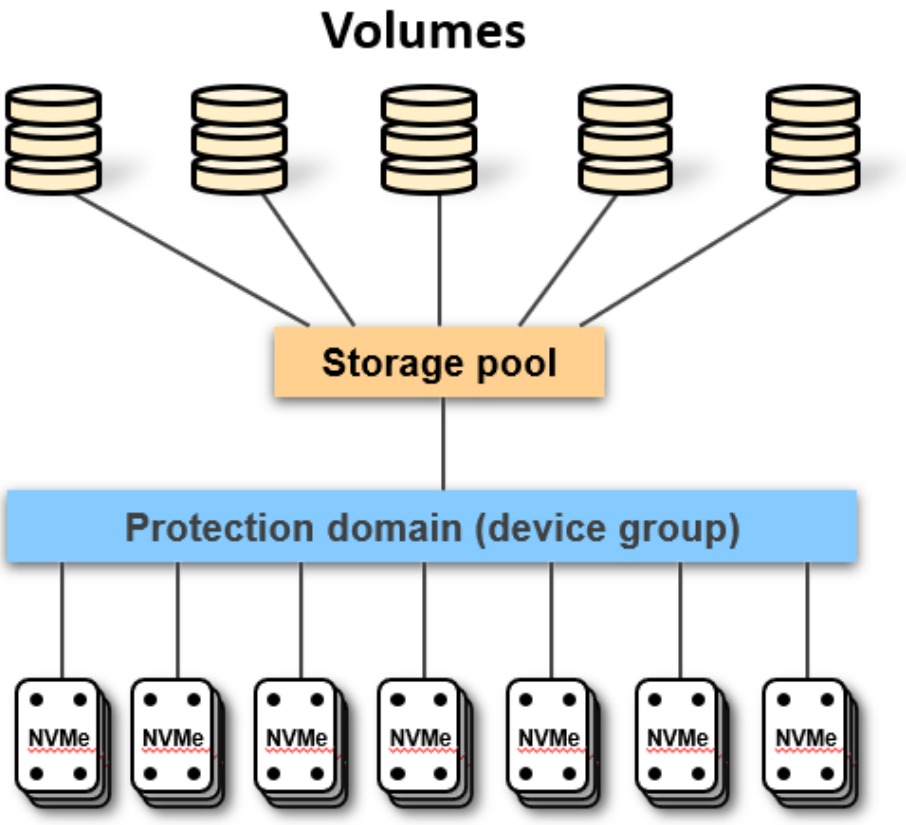


Figure 8. Volumes within the PowerFlex architecture

Only thin volumes are supported in the PowerFlex erasure coding architecture. Capacity is “on reserve,” but not allocated until used. This policy enables more flexibility in provisioning. Whereas thick capacity is limited to available capacity, thin capacity provisioning can be oversubscribed, as follows:

- Maximum thin capacity provisioning = 5 * (total raw capacity - used capacity)
- When capacity usage reaches the level where it may cause I/O errors, alerts are generated. At certain higher capacity levels, volumes can no longer be created.

Define volume names according to PowerFlex object naming conventions.

Volume copies

Volume copies are pointer-based objects that provide point-in-time copies of data that are stored in snapshots or thin clones for local data protection. These volume copies provide a method of recovery for data that has been corrupted or accidentally deleted.

PowerFlex uses redirect-on-write technology to copy data from one volume to another. This technology ensures that system space is used optimally and reduces the management burden by never requiring administrators to designate protection space.

Volume copies can be created either manually or automatically within a PowerFlex system. They are considered write-order and crash-consistent.

PowerFlex supports two types of volume copies: snapshots, which are read-only copies, and thin clones, which are read/write copies of the data.

Snapshots

Snapshots are read-only copies of volumes that are used for local data protection. They provide a method of recovery for data that has been corrupted or accidentally deleted.

Snapshots are pointer-based objects that provide point-in-time copies of data that are stored in volumes or thin clones. They can be created either manually or automatically within a PowerFlex system and are considered write-order and crash-consistent.

PowerFlex allows you to take up to 1022 snapshots per volume of existing volumes. The snapshots are thin provisioned and are extremely quick.

Because snapshots are read-only objects and cannot be modified, this allows snapshots to serve as ransomware protection if PowerFlex system data is compromised.

Snapshots cannot be mapped to the hosts. Once a snapshot is generated, it becomes a new, unmapped volume in the system. Volumes can be restored from snapshots, and thin clones (writable snapshots) can be created from them.

You can delete snapshots, but cannot resize them.

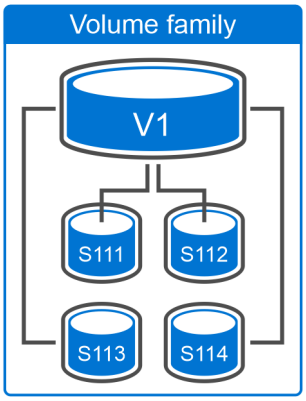


Figure 9. Snapshot operations

Keep the following considerations in mind when using snapshots:

- Because snapshots are not full copies of the original data, they should not be relied upon as a backup or as the disaster recovery solution.
- A write-order, crash-consistent snapshot is not considered application consistent, because the snapshot might not be a full representation of the application dataset at that point in time.
- Typically, a host or client caches data with the intent to write it to the storage resource. Cached data is not available within the storage when a snapshot is taken.
- Snapshots also consume overall system storage capacity to preserve the point in time. Ensure that the storage pool has enough capacity to accommodate snapshots.
- Removing a volume also removes all its direct snapshots.
- Dell Technologies does not recommend trimming volumes that contain snapshots, because the logical capacity may not shrink.

Thin clone

A thin clone is a read/write copy of a volume or a snapshot that allows you to change its content through writes.

Thin clones are thin copies of the object from which they were created. As with snapshots, thin clones are thin, pointer-based objects that use redirect-on-write technology that provides immediate access to the data contained in the source of the thin clone. Thin clones are not full copies of the original source. Because they share data blocks with the parent resource, they should not be used for disaster recovery scenarios.

The following figure shows an example of a thin clone that is created from a supported resource. When initially created, the thin clone shares all blocks with the resource from which it was created. Due to redirect-on-write technology, as new writes to the original resource or the thin clone are made, new space is consumed, and the original data remains until it is no longer in use.

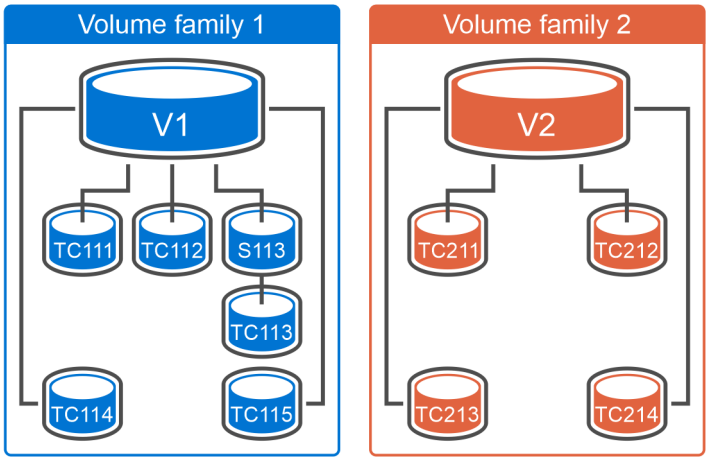


Figure 10. Thin clone operations

Thin clones can be mapped to the host and be used like normal volumes. If you want to map and use the snapshot volume, you must create a thin clone of the snapshot and map it. If you want to delete the parent volume, thin clones are unaffected.

NOTE: If you select **Delete volume with all of its thin clones** during a volume deletion operation, all the thin clones are deleted along with the volume.

Volume families

A volume family consists of the root volume and all its descendant volumes and snapshots.

A volume family is the structure that is comprised of the volume, snapshots, and thin clones resulting from that volume. It is a tree spanning from the source volume at its root, whose descendants are:

- Snapshots of the volume
- Thin clone of the snapshot
- Thin clone of the volume
- Thin clone of the thin clone
- Snapshot of the thin clone

In the following diagram of a volume family:

- TC111 & TC112 are thin clones of V1 (volume1)
- S111 is a snapshot of V1.
- TC211 is a thin clone of snapshot S111.
- S211 is a snapshot of thin clone TC211.
- S311 is a snapshot of thin clone TC112.
- TC311 is a thin clone of snapshot S311.
- TC411 is a thin clone of thin clone TC111.

Together, V1, S1xx, TC1xx, S2xx, TC2xx, S3xx, TC3xx, and TC4xx are the volume family of V1.

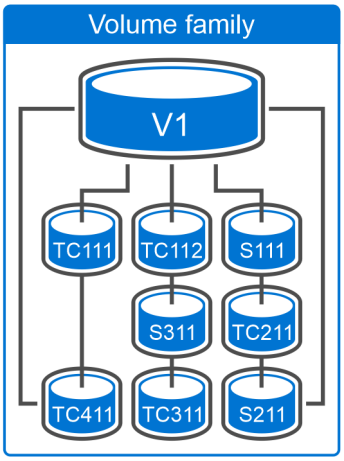


Figure 11. Volume family structure

Snapshot policies

Snapshot policies enable you to define policies where you can configure the number of snapshots to take at a given time for one or more volumes.

Snapshots are taken according to a customer-defined rule. You can define the time interval in between two rounds of snapshots, and the number of snapshots to retain in a multilevel structure. For example, take snapshots every x minutes, hours, days, or weeks. There are one to six levels, with the first level having the most frequent snapshots.

For example, you might define a rule to take snapshots every 60 minutes, with the following retention levels:

- 24 snapshots
- Seven snapshots
- Four snapshots

After the parameters for the policy are defined, the source volume is selected to add to the snapshot policy. You can add multiple source volumes to a snapshot policy, but only a single policy per source volume is allowed. Only one volume per volume family may be used as a source volume of a policy (any policy). When you remove the source volume from the policy, you must choose how to handle snapshots.

You can optionally specify that the snapshots created by the snapshot policy are secured. When snapshots are secured, the snapshots created by the policy cannot be modified or deleted before the end of the snapshot retention period. Also, the parent volume of the snapshots cannot be deleted during the snapshot retention period.

Snapshots that are created by the policy are referred to as auto snapshots. PowerFlex Manager displays the snapshot policy for each snapshot, which also indicates that it is an autosnapshot.

- If the source volume has no auto snapshots, you cannot unassign the source volume from the snapshot policy. However, you can remove auto snapshots in PowerFlex Manager.
- If the source volume has auto snapshots, but none of them are locked, you are prompted to confirm that you would like to delete all auto snapshots. If any of the auto snapshots are locked, the locked auto snapshots are detached from the snapshot policy, but not deleted, and you must remove the auto snapshots manually.

5

System hardware

There are four types of PowerFlex nodes: PowerFlex storage-only nodes, PowerFlex compute-only nodes, PowerFlex management nodes, and VMware NSX Edge nodes.

Storage-providing nodes

PowerFlex storage-only nodes are based on Dell PowerEdge R660 and R760 servers.

PowerFlex storage-only nodes

PowerFlex storage-only nodes are designed to provide storage capacity but no compute power to the compute cluster. Only the device gateway target and PowerFlex data server (storage node components) runs on PowerFlex storage-only nodes. PowerFlex storage-only nodes run an embedded operating system and do not require any VMware ESXi license. PowerFlex storage-only nodes have the ability to add additional storage capacity to a PowerFlex cluster without additional compute power.

Storage-consuming nodes

PowerFlex compute-only nodes

PowerFlex compute-only nodes are based on PowerEdge R670, R770, R6725, R7725, R660, R760, R860, R6625, and R7625 servers. The PowerFlex compute-only node enables you to deploy PowerFlex in a two-layer architecture that delivers ultimate flexibility when it comes to independently scaling compute and storage resources. The PowerFlex SDC software component is installed on PowerFlex compute-only nodes.

PowerFlex management cluster

PowerFlex management nodes are based on the Dell PowerEdge R660 servers.

PowerFlex management nodes use PowerFlex to provide a reliable, and highly available storage cluster for the management plane. PowerFlex appliance supports a multi-node PowerFlex management cluster.

PowerFlex management cluster (KVM) is a minimum of three nodes. The management cluster runs a dedicated instance of PowerFlex Manager to manage it.

6

Management control plane

The following table describes the supported deployment options for the PowerFlex management platform:

Table 9. PowerFlex management platform deployment options

Deployment offering	PowerFlex software	PowerFlex appliance	PowerFlex rack	PowerFlex custom node	Description
PowerFlex management cluster	Not supported	Supported	Supported	Not supported	A KVM with a minimum of three nodes. The management cluster runs a dedicated instance of PowerFlex Manager to manage it. Both the PowerFlex and PowerFlex Manager versions running on the management cluster are 4.x. A second instance of PowerFlex Manager, manages the PowerFlex storage clusters.
Single VMware ESXi management -based node	Not supported	Supported	Not supported	Supported	VMware ESXi-based single node with local RAID 5 storage solution.
Customer-provided infrastructure for PowerFlex management (ESXi)	Supported	Supported	Not supported	Supported	Allows you to supply a hypervisor to host the management virtual machines.
Customer-provided infrastructure for PowerFlex management (KVM/Linux)	Supported	Supported	Not supported	Supported	Allows you to supply a hypervisor to host the management virtual machines.
Linux-based customer-provided infrastructure for PowerFlex management platform. This is not a standard offering and requires special approval.	Supported	Supported	Not supported	Supported	Allows you to supply three bare-metal Linux-based servers to host the PowerFlex management platform.
Integrated PowerFlex management (non-KVM)	Supported	Not supported	Not supported	Supported	The PowerFlex storage nodes host the management instances of Kubernetes clusters and containers in a co-residency configuration.

The management network connection consists of the following:

- PowerFlex management platform - software management and orchestration stack for PowerFlex. It is implemented on the PowerFlex management cluster. It includes the container environment running on physical or virtual Linux instances, and containers that provide services.
- PowerFlex Manager - provides IT operations management for PowerFlex appliance. It increases efficiency by reducing time-consuming manual tasks that are required to manage system operations. Use PowerFlex Manager to deploy and manage new and existing PowerFlex appliance environments.

PowerFlex Manager discovers, deploys, and operates the PowerFlex appliance by using resources, templates, and resource groups.

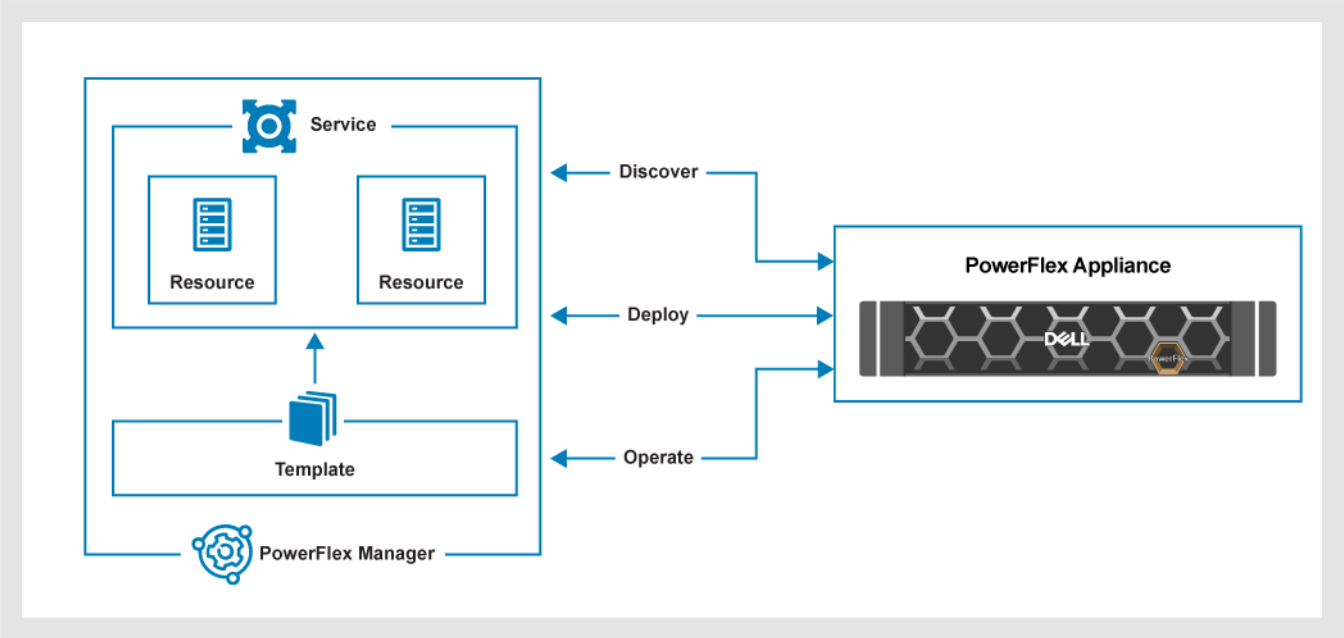


Figure 12. Management control plane

For more information on key PowerFlex Manager terminology, see the *Dell PowerFlex Glossary*.

PowerFlex Manager offers the following features:

- Resource discovery, inventory, and management
- Simplified and efficient day-to-day operations
- Management of block storage objects
- Creating template-based configurations for consistent and secure deployment of large numbers of compute, storage and network resources
- Built-in role-based authorization and identity management
- Comprehensive health alerting, monitoring, reporting and dashboards
- End-to-end automated life cycle management
- Lifecycle compliance management and reporting
- VMware vCenter - Is used for orchestrations, management, monitoring and reporting of virtual compute resources.
- Secure connect gateway - Is an enterprise monitoring technology that is delivered as an appliance and a stand-alone application. It monitors your devices and proactively detects hardware issues that may occur.
- Policy manager - policy manager is a device access management technology that is delivered as a virtual appliance.

For an in-depth overview of PowerFlex Manager, see the *PowerFlex Manager User Guide*.

7

Security considerations

Enterprises have many reasons for encrypting data, including addressing regulatory compliance, protecting against theft of customer data, and sensitive intellectual property.

PowerFlex appliance offers numerous built-in security features and capabilities cross multiple security domains to help you meet security and compliance requirements. This section provides a summary of the PowerFlex appliance security features by security domains.

Asset management

- PowerFlex Manager simplifies asset discovery and system resources inventory management
- Resource deployment services template and resource tagging allows you to efficiently deploy a complex environment with consistency

Identity authentication and authorization

PowerFlex appliance architecture offers built-in security controls to meet authentication and authorization needs. Key security controls are:

- LDAP/active directory integration
- Role-based access control (RBAC)
- RSA SecurID MFA option (using key cloak)

Data confidentiality

Confidentiality is one of the key pillars of the confidentiality, integrity, and availability (CIA) security triad. PowerFlex offers data confidentiality through data at rest encryption (D@RE).

PowerFlex supports CloudLink D@RE-based encryption. CloudLink offers the following encryption methods:

- CloudLink software encryption: CloudLink can be integrated with PowerFlex to provide software-based encryption for PowerFlex storage nodes that is transparent to the features and operation of the PowerFlex solution. CloudLink uses dm-crypt, a native Linux encryption package, to secure storage node devices. A proven high-performance volume encryption solution, dm-crypt is widely implemented for Linux machines.
- Hardware-based FIPS 140-2 compliant data at rest encryption: For hardware-based D@RE, you can select from a supported list of self- encrypting drives (SEDs) that meet your business needs, using integrated CloudLink for key management.

CloudLink encrypts the storage node devices with unique keys that are controlled by enterprise security administrators. CloudLink Center provides centralized, policy-based management for these keys, enabling single-screen security monitoring and management across one or more PowerFlex deployments.

System trust

PowerFlex appliance is built with PowerFlex nodes, which are PowerEdge servers. The PowerFlex nodes inherit all the cutting-edge cyber-resiliency and security features such as:

- An immutable silicon-based root of trust to securely boot iDRAC, BIOS and firmware
- Virtual lock for preventing server configuration/firmware changes and drift detection
- Rapid recovery to a trusted image when authentication fails
- Rollback to known good firmware version if firmware is compromised
- Secure system erase internal server storage devices including HDD, SSD, and NVMe drives

- Industry leading secure supply chain
- PowerFlex software integrity check

Network security

PowerFlex appliance not only offers built-in access and aggregation or leaf-spine network topology but also incorporates many advanced security features that are available with Cisco Nexus and Dell PowerSwitch switches. These security features help you to protect network against data loss or compromise resulting from intentional attacks and from unintended but damaging actions made by well-meaning network users. Key security features include:

- Network segmentation with, ACL, firewall, and VLAN
- TACACS+ Security Protocols support
- LDAP authentication and authorization support
- Role-based access control (RBAC) to control and limit access to operations on the Cisco NX-OS device
- Authentication, authorization, and accounting (AAA) an architectural framework support
- Access control list (ACL) support. IP ACLs, MAC ACL and VACL are available options to filter traffic based on IPv4 addresses, MAC address in the packet header, and VLAN routing
- Simple Certificate Enrollment Protocol (SCEP) support
- Dynamic ARP inspection, DHCP Snooping, key chain management, and control plane policing can used to further harden the security

Auditing and accountability

Audit and accountability's primary objectives are to maintain a record of system activities, and provide the ability to establish individual accountability, detect system anomalies, reconstruct system events using audit logs and records. PowerFlex appliance creates and retains system audit logs, event logs and alert records to that can be used for monitoring, trend and behavior analysis, incident investigation, and reporting of unlawful or unauthorized system activities.